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# BookHive · E2E QA Summary

An automated end-to-end test suite, delivered with **spriteCloud** methodology.

***Test your software, not your reputation.***

Full six-stage QA pipeline executed against BookHive's bookstore + peer-to-peer marketplace: journey mapping, onboarding, coverage expansion, and adversarial bug discovery — from zero tests to a green suite and a critical bug, in one session.

22 April 2026 · branch: experimental/no-tests · spritecloud.com

# Project Overview

BookHive is a full-stack bookstore + peer-to-peer marketplace purpose-built as a target for test automation — stable `data-testid` selectors, idempotent seed/reset endpoints, a 50-book fixture, two test users.

## Application under test

**Frontend:** React 18 SPA (Vite) → Nginx on :7547
**Backend:** Spring Boot 3.2, Java 17 on :8080
**Database:** MongoDB 7
**Domain:** 50 books across 6 genres, cart + checkout with a 10-minute return window, peer-to-peer marketplace.

## Automation stack

@civitas-cerebrum/element-interactions ^0.2.8
@civitas-cerebrum/element-repository ^0.1.8
@playwright/test ^1.52

TypeScript 5.6 · Playwright MCP for discovery · Docker Compose test stack. The Steps API keeps raw selectors out of test code.

## Starting point

Branch `experimental/no-tests` — the prior E2E suite and QA CI were intentionally removed in commit `658fa66`. Session began with zero automated tests.

## Final state

44/44 PASSING main suite · 7/7 FAILING bug-discovery repro · **10** commits · **6** findings in `docs/e2e/bug-discovery-report.md`.

# What Was Built

Six-stage pipeline executed end-to-end in a single session, each stage gated and committed before the next.

**44**
PASSING TESTS

**12 / 12**
JOURNEYS COVERED

**11**
PAGES IN REPO

**99**
SELECTORS

**6**
BUGS FOUND

### Suite health

Main suite runs in ~1 min against the live Docker Compose stack. Every action is API-verified (balance/status/stock deltas) — tests don't stop at "button was clicked".

### Coverage depth

Every P0 journey covered with happy path AND at least one error/edge case. Every page in the app reached by at least one test. Mobile P0 walked end-to-end at 375×812.

### Bugs

1 Critical financial-integrity exploit confirmed live via the UI. 1 Medium UX defect, 3 Low API-validation gaps, 1 DOM-only. All captured as failing reproduction tests.

# Session Timeline

All six stages of the spriteCloud QA workflow executed back-to-back, with an approval gate at each stage boundary.

JOURNEY MAPPING

**App discovered via Playwright MCP; 12 user journeys prioritised P0–P3**

Live-MCP exploration of 11 pages. `app-context.md` + sentinel-bearing `journey-map.md` emitted for approval.

STAGES 1–4 · ONBOARDING

**12 specs covering every mapped journey — 19/19 passing**

P0 (checkout, return, signup, buy-listing) · P1 (browse, create/cancel listing, cart mgmt) · P2/P3 (guest detail, theme, logout, invalid login).

STAGE 5 · CYCLE 1

**Expansion — mobile, navigation, cart extras, perf (+13 tests)**

Mobile P0 checkout · all six sidebar genre filters parameterised · pagination edges · cart-badge count · home perf smoke under 5s.

STAGE 5 · CYCLE 2

**Expansion — API guardrails, orders list, stock, signup-buy (+12 tests)**

Every protected endpoint rejects guests · multi-status orders list · peer checkout decrements stock visibly · signup-then-buy end-to-end.

STAGE 6 · BUG DISCOVERY

**Adversarial probing — 1 Critical · 1 Medium · 3 Low · 1 No-impact**

Ghost-order race confirmed exploitable end-to-end ($100 → $112.99). Five additional findings with failing reproduction tests checked in.

# Journey Coverage

12 user journeys mapped from the live app, prioritised by business impact. Every P0 has full-path + error-state coverage; every P1 has its core path; every P2/P3 has at least a smoke check.

| JOURNEY | PRIORITY | SPEC | COVERAGE |
|---|---|---|---|
| J1 · Checkout a book (happy path + insufficient balance) | P0 | `p0-j1-checkout` | 100% |
| J2 · Return within the 10-minute window | P0 | `p0-j2-return` | 100% |
| J3 · Signup + starter balance (happy + duplicate email) | P0 | `p0-j3-signup` | 100% |
| J4 · Buy a marketplace listing (+ own-listing no-buy) | P0 | `p0-j4-buy-listing` | 100% |
| J5 · Browse / filter / search / paginate | P1 | `p1-j5-browse` | 100% |
| J6 · Create a marketplace listing | P1 | `p1-j6-create-listing` | 100% |
| J7 · Cancel own listing | P1 | `p1-j7-cancel-listing` | 100% |
| J8 · Cart qty / remove / clear | P1 | `p1-j8-cart` | 100% |
| J9 · Guest views book detail | P2 | `p2-j9-guest-detail` | 100% |
| J10 · Theme toggle + responsive layout | P2 | `p2-j10-theme-responsive` | 100% |
| J11 · Logout clears session | P3 | `p3-j11-logout` | 100% |
| J12 · Invalid login surfaces error | P3 | `p3-j12-invalid-login` | 100% |

# Element Repository

11 pages × 99 stable selectors — all `data-testid`-backed, zero raw selectors in test code. Test files reference `('elementName', 'PageName')` strings; the repository resolves each to a Playwright locator at runtime.

| PAGE | ELEMENTS | TESTED BY |
|---|---|---|
| `Shell` (sidebar + topbar) | 24 | Every spec — nav assertions and user-balance checks |
| `HomePage` | 21 | J5 · expand-navigation · expand-cart-extras · expand-performance |
| `BookDetailPage` | 8 | J1 · J9 · expand-mobile-checkout · expand-stock-and-signup-buy |
| `LoginPage` | 6 | J1, J2, J4, J6, J7, J8, J11, J12 · most expansion specs |
| `SignupPage` | 7 | J3 · expand-stock-and-signup-buy · BUG-001 repro |
| `CartPage` | 8 | J1 · J8 · expand-cart-extras · expand-mobile-checkout |
| `OrdersPage` | 3 | expand-orders-list |
| `OrderDetailPage` | 6 | J1 · J2 · expand-mobile-checkout · expand-cart-extras |
| `MarketplacePage` | 4 | J4 · J7 · J6 · expand-api-guardrails |
| `CreateListingPage` | 5 | J6 · BUG-004 + BUG-005 repro |
| `ProfilePage` | 7 | J3 · J6 · J7 |

# Bug Discovery Findings

Stage-6 adversarial probing — element boundaries, flow races, context-derived. Every finding has a failing reproduction test under `tests/bug-discovery/` that turns green when the bug is fixed.

| ID | TITLE | SEVERITY | VISIBILITY |
|---|---|---|---|
| **BUG-003** | Race on `POST /api/orders` creates ghost orders; returning them credits money never charged. Confirmed $100 → $112.99 exploit. | CRITICAL | User-visible |
| **BUG-001** | Signup with a short password shows a generic "unexpected error" (unhandled 500) instead of a field-level validation message. | MEDIUM | User-visible |
| **BUG-002** | `POST /api/cart/items` returns HTTP 500 when `quantity <= 0` instead of 400. | LOW | API-only |
| **BUG-004** | `POST /api/marketplace/listings` accepts arbitrary `condition` strings — no backend allow-list. | LOW | API-only |
| **BUG-005** | `POST /api/marketplace/listings` returns HTTP 500 on zero / negative price or overlong condition. | LOW | API-only |
| **BUG-006** | Guest page loads log a 403 from `/api/auth/me` — devtools noise only; UI handles correctly. | NO IMPACT | DOM-only |

**BUG-003 — the headline**

Triple-clicking Checkout on a 1-book cart creates 3 orders. Only one is real; the rest are "ghosts" showing identical contents. Stock and balance each decrement once, but **returning a ghost order still credits $12.99** — net positive per exploit. Two failing repro tests assert correct behaviour.

# What Ships With the Suite

Tests are the deliverable — but the supporting artifacts are what make the suite survive the next change in the app.

### 📍 Journey map

`docs/journey-map.md` — 12 prioritised journeys, sentinel-bearing for resumability. Coverage checkpoint updated per cycle.

### 📚 App context

`docs/app-context.md` — testids, state transitions, gated areas. Starting knowledge base for the next session.

### 📖 Plain-English scenarios

`docs/e2e-test-scenarios.md` — 45 scenarios as a human tester would write them; reviewable by non-technical stakeholders.

### 🔧 API + UI helpers

`helpers/api.ts` seeds data authoritatively; tests drive the UI only for UX assertions. State setup uses the API for speed.

### 🐛 Bug reproduction suite

`tests/bug-discovery/` + dedicated config. Runs in isolation; failing tests turn green when bugs are fixed.

### 📸 Visual evidence

`docs/screenshots/` — BUG-003 ($112.99 balance after refund) + BUG-001 (generic-error signup).

### Run commands

**Main suite (44):** `npx playwright test` · **Bug repro (7 failing):** `npx playwright test --config=playwright.bug-discovery.config.ts`

**Reset DB:** `curl -X POST http://localhost:8080/api/reset`

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# A full E2E suite — and a critical bug — in one session

Everything committed on `experimental/no-tests` · reproducible with `docker compose up` + `npx playwright test`.

| | |
|---|---|
| @civitas-cerebrum/element-interactions | 0.2.8 |
| @civitas-cerebrum/element-repository | 0.1.8 |
| @playwright/test | 1.52 |

**44** TESTS GREEN

**12** JOURNEYS

**99** SELECTORS

**1** CRITICAL BUG

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